



Solving the pentagon equation for unitary solutions yields a  $F$  associator

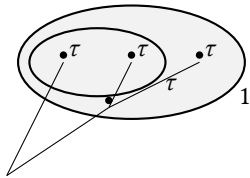
$$F_{\tau}^{\tau\tau\tau} : \mathbb{C} \left[ \begin{array}{c} \tau \quad \tau \quad \tau \\ \swarrow \quad \downarrow \quad \searrow \\ 1 \end{array} \right] \oplus \mathbb{C} \left[ \begin{array}{c} \tau \quad \tau \quad \tau \\ \swarrow \quad \searrow \quad \downarrow \\ \tau \end{array} \right] \xrightarrow{\cong} \mathbb{C} \left[ \begin{array}{c} \tau \quad \tau \quad \tau \\ \swarrow \quad \downarrow \quad \swarrow \\ \tau \end{array} \right] \oplus \mathbb{C} \left[ \begin{array}{c} \tau \quad \tau \quad \tau \\ \swarrow \quad \swarrow \quad \downarrow \\ \tau \end{array} \right]$$

given by the basis expansions (well rather these are given by the solution. But the solution exists because of the linearity of the hom spaces.)

$$\left[ \begin{array}{c} \tau \quad \tau \quad \tau \\ \swarrow \quad \downarrow \quad \searrow \\ 1 \end{array} \right] =? \left[ \begin{array}{c} \tau \quad \tau \quad \tau \\ \swarrow \quad \downarrow \quad \swarrow \\ \tau \end{array} \right] \oplus? \left[ \begin{array}{c} \tau \quad \tau \quad \tau \\ \swarrow \quad \swarrow \quad \downarrow \\ \tau \end{array} \right]$$

$$\text{Hom}(A \otimes B \otimes C, D) = \text{span} \left\{ \begin{array}{c} A \quad B \quad C \\ \swarrow \quad \downarrow \quad \searrow \\ v_1 \quad i \quad v_2 \\ \searrow \\ D \end{array} \right\}$$

$$\begin{array}{c} A \quad B \quad C \\ \swarrow \quad \downarrow \quad \searrow \\ \mu \quad v \quad e \\ \searrow \\ D \end{array} = \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(e\mu\nu)}^{(f\alpha\beta)} \begin{array}{c} A \quad B \quad C \\ \swarrow \quad \downarrow \quad \searrow \\ v_1 \quad i \quad v_2 \\ \searrow \\ D \end{array}$$

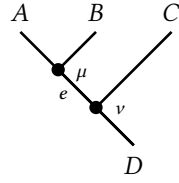


$$\begin{array}{ccccc}
(A \otimes B) \otimes C & \xrightarrow{\alpha_{ABC}} & A \otimes (B \otimes C) \\
\downarrow \text{Hom}(-, D) & \text{Yoneda-lifting} & \downarrow \text{Hom}(-, D) \\
\text{Hom}((A \otimes B) \otimes C, D) & \cong & \text{Hom}(A \otimes (B \otimes C), D) \\
\downarrow \text{semisimplicity} & \text{semisimplicity} & \downarrow \text{semisimplicity} \\
\bigoplus_c \text{Hom}(A \otimes B, c) \otimes \text{Hom}(c \otimes C, D) & \cong & \bigoplus_{d \in \text{Irr}(\text{Cat})} \text{Hom}(A \otimes d, D) \otimes \text{Hom}(B \otimes C, d) \\
\downarrow \text{choice of basis} & \text{choice of basis} & \downarrow \text{choice of basis} \\
\bigoplus_c V_c^{AB} \otimes V_D^{cC} & & \bigoplus_c V_D^{Ac} \otimes V_c^{BC} \\
\downarrow \text{choice of basis} & & \downarrow \text{choice of basis} \\
|(A, B); c; \mu\rangle \otimes |c, C; D; \nu\rangle & \xlongequal{\hspace{1cm}} & \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(c\mu\nu)}^{(f\alpha\beta)} |A, f; D; \beta\rangle \otimes |B, C; f; \alpha\rangle
\end{array}$$

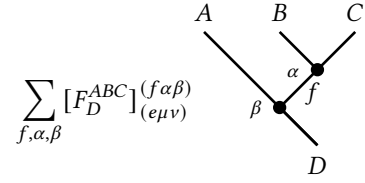
Let  $f : E \rightarrow A \otimes B$  and  $g : D \rightarrow E \otimes C$ .

$$\begin{array}{ccc}
(A \otimes B) \otimes C & \xrightarrow[\sim]{\alpha_{ABC}} & A \otimes (B \otimes C) \\
\downarrow \text{Hom}(-, D) & & \downarrow \text{Hom}(-, D) \\
\text{Hom}((A \otimes B) \otimes C, D) & \xrightarrow[\text{associator induces isomorphisms under yoneda}]{\alpha_{A,B,C} \sim} & \text{Hom}(A \otimes (B \otimes C), D) \\
\downarrow \text{semisimplicity} \parallel & & \downarrow \text{semisimplicity} \parallel \\
\bigoplus_{c \in \text{Irr}(\text{CAT})} \text{Hom}(A \otimes B, c) \otimes \text{Hom}(c \otimes C, D) & \cong & \bigoplus_{d \in \text{Irr}(\text{Cat})} \text{Hom}(A \otimes d, D) \otimes \text{Hom}(B \otimes C, d) \\
\parallel & & \parallel \\
\bigoplus_c V_c^{AB} \otimes V_D^{cC} & \cong & \bigoplus_c V_D^{Ac} \otimes V_c^{BC} \\
\mathfrak{m} & & \mathfrak{m} \\
|(A, B); c; \mu\rangle \otimes |c, C; D; \nu\rangle & \xrightarrow[\text{6j-symbols}]{F_D^{ABC}} & \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(c\mu\nu)}^{(f\alpha\beta)} |A, f; D; \beta\rangle \otimes |B, C; f; \alpha\rangle
\end{array}$$

$\mu \in 1, \dots, N_c^{AB} \quad \nu \in 1, \dots, N^{cC, D}$



=



$$\begin{array}{ccc}
(A \otimes B) \otimes C & \xrightarrow[\sim]{\alpha_{ABC}} & A \otimes (B \otimes C) \\
\downarrow \text{Hom}(-, D) & & \downarrow \text{Hom}(-, D) \\
\text{Hom}((A \otimes B) \otimes C, D) & \xrightarrow[\sim]{\alpha_{A,B,C}} & \text{Hom}(A \otimes (B \otimes C), D) \\
\downarrow \text{semisimplicity} \quad \parallel & \text{associator induces isomorphisms under yoneda} & \downarrow \text{semisimplicity} \quad \parallel \\
\bigoplus_{c \in \text{Irr}(\text{CAT})} \text{Hom}(A \otimes B, c) \otimes \text{Hom}(c \otimes C, D) & \cong & \bigoplus_{d \in \text{Irr}(\text{Cat})} \text{Hom}(A \otimes d, D) \otimes \text{Hom}(B \otimes C, d) \\
\parallel & & \parallel \\
\bigoplus_e V_e^{AB} \otimes V_D^{eC} & \cong & \bigoplus_f V_D^{Af} \otimes V_f^{BC} \\
\cap & & \cap \\
|(A, B); e; \mu\rangle \otimes |e, C; D; v\rangle & \xrightarrow[\text{6j-symbols}]{F_D^{ABC}} & \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(e\mu\nu)}^{(f\alpha\beta)} |A, f; D; \beta\rangle \otimes |B, C; f; \alpha\rangle
\end{array}$$

